

#Aviation Meteorology



Chapter One: Introduction

Objectives

- To Be familiar with basic concept of Aviation Meteorology
- To Know the Nature and scope of Aviation Meteorology
- To be familiar with ICAO definitions of different terms
- To know *The Basic Forces Acting On Aircraft*



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The Basic Concept Of Aviation Meteorology

- Aviation meteorology, also known as aeronautical meteorology or simply aviamet, is a specialized branch of meteorology that focuses on providing weather information and services to support safe and efficient aviation operations.

- The basic concept of aviation meteorology is to provide accurate and reliable weather information to ensure **safe** and efficient **flight** operations.

- This information includes **weather reports**, **forecasts**, and advisories that cover a wide range of factors, such as wind speed and direction, visibility, cloud cover, turbulence, thunderstorms, icing conditions, and other weather-related hazards.
- One of the main challenges of aviation meteorology is to provide **timely** and **accurate** weather information that can help pilots make informed decisions about flight operations.

- This is particularly important during **takeoff** and **landing** when weather conditions can change rapidly and have a significant impact on flight safety.
- To meet this challenge, aviation meteorologists use a variety of **tools** and **technologies** to collect and analyze weather data, including **satellite imagery**, **radar**, and **surface observations**.
- They also use computer models to forecast weather conditions based on **historical data** and **current** atmospheric conditions.

- Another important aspect of aviation meteorology is to provide tailored weather information to specific users, such as pilots, air traffic controllers, and airport operators.
- For example, pilots need information about wind speed and direction to plan their flight routes and altitudes, while air traffic controllers need information about weather conditions to manage traffic flow and ensure safe operations.

- In summary, aviation meteorology is a critical component of the aviation industry that provides **crucial weather information and forecasts** to ensure safe and efficient flight operations.
- It requires specialized **knowledge, skills, and technologies** to **collect, analyze, and communicate** weather information to specific users in a timely and accurate manner.

- *Aviation meteorologists* provide weather information to airline operations, flight dispatchers, and pilots.
- The meteorologist must determine both the current and forecasted weather conditions for all altitudes.
- These conditions include the wind's direction and speed, cloud cover, visibility, and precipitation.

- Aviation weather reports differ from other weather reports because they *consider what conditions may affect flights*.
- For example, if heavy turbulence is likely at one altitude, the pilot needs to know what altitude to climb to, avoiding the turbulence.
- Aviation meteorologists rely on radar, computers, weather station information, and other tools to compile their weather reports.

Nature and Scope of Aviation Meteorology

- Aviation meteorology is a specialized field of meteorology that deals with **weather-related** issues in aviation.
- It provides crucial information to aviation professionals to ensure the safety of **passengers** and **crew** during flights.

- The nature of aviation meteorology is centered around the collection, analysis, and communication of weather data.
- The data includes current weather **observations**, **weather forecasts**, and **advisories**.

- The data can be analyze using **computer models** and **statistical methods** to forecast weather conditions.
- The communication of the data is done through various channels, such as **aviation weather reports**, **aviation weather forecasts**, and **aviation weather warnings**.
- The scope of aviation meteorology is vast and covers several aspects of **aviation** operations.

- One of the main areas of scope is to provide weather *forecasts for the aviation industry*.
- The forecasts help aviation professionals plan and execute their operations in a safe and efficient manner.
- Weather forecasts include information about wind speed and direction, visibility, cloud cover, turbulence, thunderstorms, icing conditions, and other weather-related hazards.

- The meteorologists also provide information about the expected onset and duration of weather events to help pilots avoid hazardous weather conditions.
- Another area of scope for aviation meteorology is to provide *weather reports* for aviation purposes.
- Weather reports are timely *updates* about the current weather conditions at an *airport* or along a *flight route*.

- The reports are essential for **flight planning** and **execution**, and they **provide information** about visibility, cloud height, cloud cover, temperature, dew point, pressure, wind direction, and speed.
- Aviation meteorologists also provide aviation weather *warnings* to aviation professionals.
- The warnings are issued to **alert pilots** and **air traffic controllers** about **hazardous** weather conditions.

- The warnings include information about thunderstorms, icing conditions, turbulence, and other weather-related hazards that can affect flight safety.
- The warnings are essential for flight planning and execution, and they help aviation professionals make informed decisions about their operations.

Aeronautical Meteorological Station

- Aeronautical meteorological station. A station designated to make observations and issue meteorological reports for use in international air navigation.

- A meteorological observations programme that meets the requirements of aviation shall be provided at aeronautical meteorological stations established, as necessary, at aerodromes and other locations of significance to international air navigation.
- The observations programme shall be as per the Technical Regulations (WMO-No. 49) Agreement between air traffic services authorities and meteorological authorities.

- The meteorological elements required to be observed at aeronautical meteorological stations include surface *wind, visibility (and RVR on all runways intended for use during periods of reduced visibility), present weather, cloud (and vertical visibility when the sky is obscured), air temperature, dewpoint temperature, atmospheric pressure values.*

- At aeronautical meteorological stations, observations are made throughout the 24 hours of each day, except as otherwise agreed between the meteorological authority, the appropriate ATS authority and the operators concerned and in accordance with regional air navigation agreements.
- Due to the importance of meteorological observations for the safe and efficient operation of aircraft, it is vital that personnel responsible for making the observations are well trained for the task.
- In order to ensure that the observers are well trained, training courses should be established and, once taken, refresher training should be given on a routine basis.

ICAO Definitions Of Terms

- ✓ routine air-report
- ✓ special air-report
- ✓ meteorological report
- ✓ aerodrome forecast
- ✓ landing forecast
- ✓ gamet area forecast
- ✓ sigmet information
- ✓ airmet information
- ✓ runway visual range

- ✓ flight information region
- ✓ terminal area
- ✓ controlled airspace
- ✓ advisory airspace
- ✓ controlled area
- ✓ control zone
- ✓ ATS route
- ✓ airway
- ✓ advisory route

Routine Air-Report: A report on the current weather conditions along a **flight route** or at an **aerodrome**. These reports are issued at fixed intervals or as requested by aircraft.

Special Air-Report: A report on the current weather conditions that **deviate significantly** from what was forecasted in the routine air-report.

Meteorological Report: A report that provides information on [weather conditions](#), including temperature, pressure, wind speed and direction, visibility, cloud cover, precipitation, and other relevant information.

Aerodrome Forecast: A forecast that provides information on [expected weather conditions at an aerodrome](#), including wind speed and direction, visibility, cloud cover, and other relevant information.

Landing Forecast: A forecast that provides information on expected weather conditions at the **time of landing**, including wind speed and direction, visibility, cloud cover, and other relevant information.

GAMET Area Forecast: A forecast that provides information on expected weather conditions over a **large geographical area**, including wind speed and direction, visibility, cloud cover, and other relevant information.

SIGMET Information: A warning issued by meteorological authorities to **alert pilots** of potentially hazardous weather conditions in a specific area.

AIRMET Information: advises of weather that may be hazardous issued by meteorological authorities to alert pilots that may affect **flight safety** in a specific area.

Runway Visual Range: Is the range over which the pilot of an aircraft on the **center line** of a runway can see the runway **surface markings** or the **lights** delineating the runway or identifying its center line.

Flight Information Region: A specific region of airspace in which a flight information service and an alerting service (ALRS) are provided.

Terminal Area: Means the Cost Center that includes all areas within the Terminals, including concourses, boarding areas, pedestrian bridges, tunnel lobbies to garages, mechanical/utility areas, lobbies, offices, Commission facilities, baggage facilities, and storage areas.

Controlled Airspace: Airspace where air traffic control services are provided to all aircraft.

Advisory Airspace: Airspace where air traffic control services are not mandatory but are available to all aircraft.

Controlled Area: A specific area of controlled airspace where air traffic control services are provided.

Control Zone: A controlled airspace extending upwards from the surface of the earth to a specified upper limit, established around an aerodrome to protect arriving and departing aircraft.

ATS Route: A specific route established for the safe and efficient flow of air traffic.

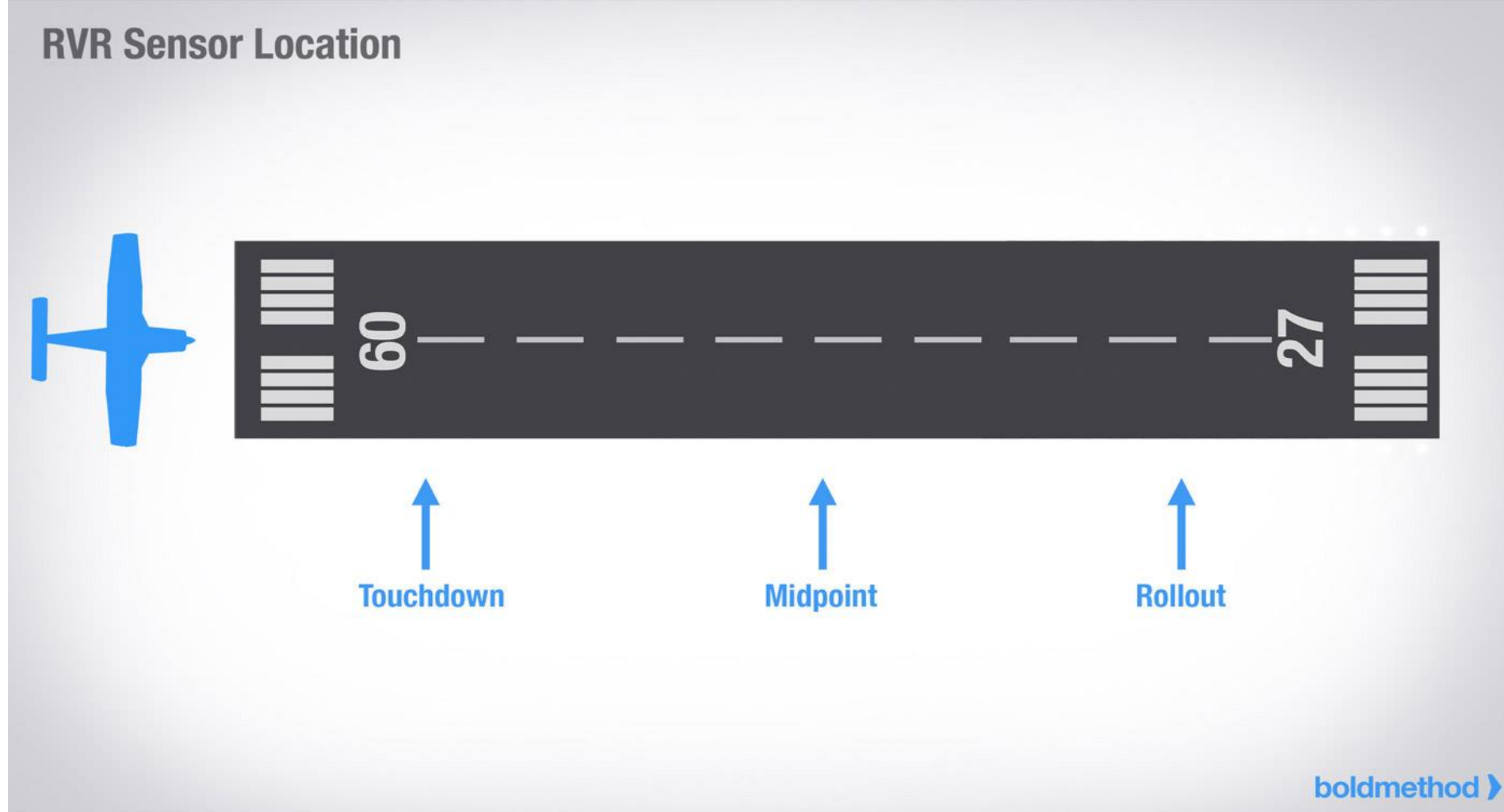
Airway: A designated path in the sky for aircraft to fly along that is defined by a specific width and altitude.

Advisory Route: A route established for the safe and efficient flow of air traffic or is a designated route along which air traffic advisory service is available.

Controlled Airspace



Runway visual range



Runway visual range



Terminal Area



GLOSSARY OF ABBREVIATIONS AND TERMS

<i>Abbreviation/ term</i>	<i>Definition</i>	<i>Abbreviation/ term</i>	<i>Definition</i>
ACC	Area Control Centre		Service for International Air Navigation (WMO-No. 49), on Aeronautical Climatology
ADS	Automatic Dependent Surveillance		
AFTN	Aeronautical Fixed Telecommunication Network—ICAO dedicated communication system	C.3.3	A chapter of the WMO <i>Technical Regulations</i> , Volume II—Meteorological Service for International Air Navigation (WMO-No. 49), on Format and Preparation of Flight Documentation
AIP	Aeronautical information publication		
AIREP	Manual (usually) report of meteorological conditions from an aircraft in flight		
AIRMET	Advice of weather that may be hazardous for, typically, general aviation (GA) aircraft flying VFR	CAA	Civil Aviation Authority—the authority within a country having overall responsibility for civil aviation matters
AIS	Aeronautical Information Service	CAeM	Commission for Aeronautical Meteorology
AMBEX	African-Indian Ocean Meteorological Bulletin Exchange	CAPPI	Constant Altitude Plan Position Indicator
AMDAR	Aircraft Meteorological Data Relay—automated aircraft reports, and also the name of a code form used to transmit such reports	CNS	Communication-Navigation-Surveillance
		CNS/ATM	Communication-Navigation-Surveillance/Air Traffic Management
Annex 3	The ICAO Annex to the Chicago Convention on Civil Aviation, which is essentially identical to Chapter C.3.1 of the WMO <i>Technical Regulations</i> , Volume II—Meteorological Service for International Air Navigation (WMO-No. 49), on Standards and Recommended Practices	FIC	Flight Information Centre
		FIR	Flight Information Region
		FL	Flight Level—equivalent to an atmospheric pressure level. For example, a flight level of 250 is that level in the atmosphere where the pressure is equal to that of the ICAO standard atmosphere at a height of 25 000 feet
		GA	General aviation

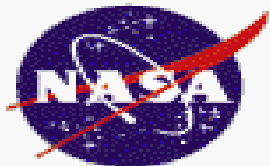
GLOSSARY OF ABBREVIATIONS AND TERMS

ANS	Air Navigation Services	GRIB	Binary, table-oriented code used to exchange gridded data
ASDAR	Aircraft-to-Satellite Data Relay—a dedicated system installed on up to 30 aircraft to provide automated reports via geostationary meteorological satellites, and now in decline in favour of AMDAR.	GTS	Global Telecommunication System
		ICAO	International Civil Aviation Organization
		IFR	Instrument Flight Rule
ASOS	Automated Surface Observing System (AWS)	ISCS	International Satellite Communications System; satellite system provided by the United States to distribute WAFS data through two satellites (Pacific and Atlantic)
ATC	Air Traffic Control		
ATIS	Automatic Terminal Information Service	IT	Information technology
ATM	Air Traffic Management	ITCZ	Intertropical convergence zone
ATS	Air Traffic Services	MDA	Multiple Discriminant Analysis (technique used in statistical forecasting)
AWS	Automatic Weather Station		
BUFR	Binary, table-oriented code used to exchange observations and forecast information such as SIGWX	METAR	Meteorological Aerodrome Report—hourly or half-hourly coded weather observation from an aerodrome
C.3.1	A chapter of the WMO <i>Technical Regulations</i> , Volume II—Meteorological Service for International Air Navigation (WMO-No. 49), on Standards and Recommended Practices. Essentially identical to ICAO Annex 3 of the Chicago Convention on Civil Aviation	Meteorological Authority	The entity which has been notified by the Member State to ICAO as being responsible for providing or arranging for the provision of meteorological service for international air navigation, in accordance with the WMO <i>Technical Regulations</i> , Volume II, [C.3.1.], Part I, paragraph 2.1.4
C.3.2	A chapter of the WMO <i>Technical Regulations</i> , Volume II—Meteorological	MOS	Model Output Statistics (statistical forecasting)

GLOSSARY OF ABBREVIATIONS AND TERMS

<i>Abbreviation/ term</i>	<i>Definition</i>	<i>Abbreviation/ term</i>	<i>Definition</i>
MOTNE	Meteorological Operational Network Europe	SARPs	Standards and Recommended Practices
MSL	Mean sea level	SDUS	Secondary Data User Stations
MWO	Meteorological watch office (responsible for issuing SIGMETs within a designated FIR or FIRs)	SIGMET	Warning of existing or forecast hazardous in-flight weather for aviation
NMC	National Meteorological Centre	SIGWX	Significant weather forecast for aviation in chart or coded form
NMHS	National Meteorological or Hydrological Service	SPECI	Special aerodrome report issued for an aerodrome when conditions have changed according to set criteria
NWP	Numerical weather prediction	SSP	Sub-satellite point
OPMET	Operational meteorological information	TAF	Terminal Aerodrome Forecast—covering a period from 9 to 24 hours in the future
PDUS	Primary Data User Station	TCAC	Tropical Cyclone Advisory Centre
PIREP	Pilot Report, often usually in plain language	TREND	Short-range forecast (2 or 3 hours depending on agreed practice) appended to a METAR
PPI	Plan position indicator	UIR	Upper flight information region
PPM	Perfect Prognosis Method (type of statistical forecasting)	UNIX	Computer operating system (similar to Linux)
QFE	Pressure at aerodrome level	VAAC	Volcanic Ash Advisory Centre—part of the ICAO Airways Volcanic Watch
QMF	Quality Management Framework	VDU	Visual Display Unit (computer monitor)
QNH	The pressure reading that an altimeter should be set to so that the geometric altitude it reads is correct when at the level of the aerodrome	VFR	Visual Flight Rules
RAN Agreement	Regional Air Navigation Agreement—agreed within regional meetings of ICAO	VHF	Very high frequency
RHI	Range height indicator	VMC	Visual meteorological conditions
ROBEX	Regional Operational Bulletin Exchange	VOLMET	Broadcast of meteorological information to aircraft in flight
ROFOR	Route forecast of upper winds and temperatures in the ROFOR code	WAFC	World Area Forecast Centre—the two centres under WAFS are London and Washington
RVR	Runway visual range	WAFS	World Area Forecast System
SADIS	Satellite Distribution System provided by the UK to distribute WAFS data	WMO	World Meteorological Organization

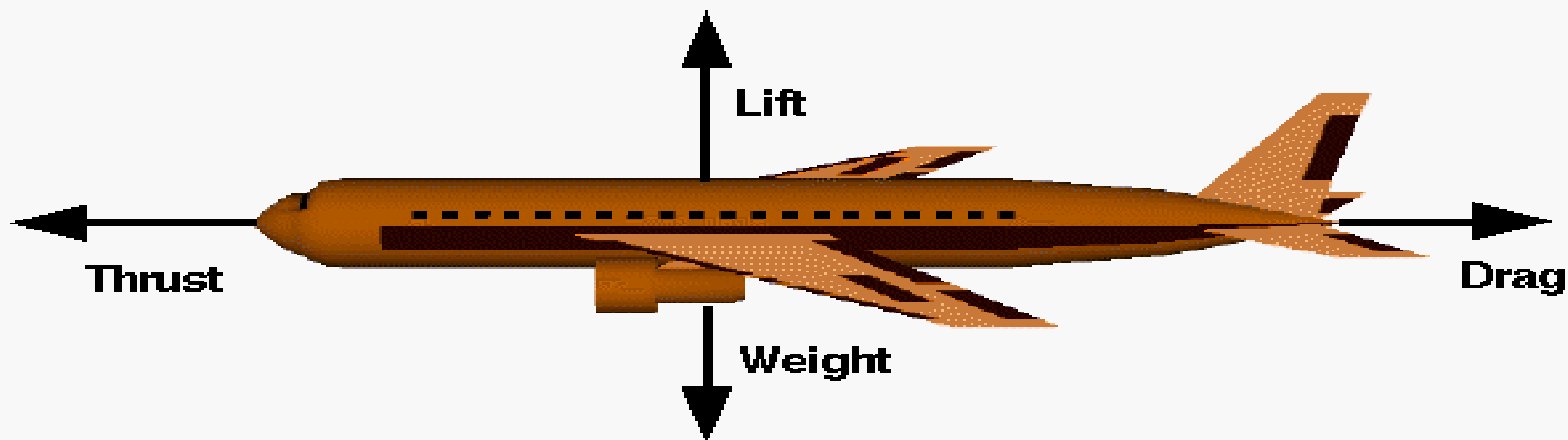
The Basic Forces Acting On Aircraft



Simplified Aircraft Motion

Unbalanced Forces

Glenn
Research
Center



Flight Condition	Effect
Lift > Weight	Plane Rises
Weight > Lift	Plane Falls
Drag > Thrust	Plane Slows
Thrust > Drag	Plane Accelerates

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Thanks !